Quantum Key Distribution-Based Security Framework for Wireless Sensor Networks: Enhancing Resilience Against Classical and Quantum Cyber Threats

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S. Senthilkumar ; R. Rajaganapathy ; K.K. Shandhosh Shree ; P. Krishnamoorthy ; L. Ramachandran ; M. Kavitha **All Authors**

Abstract:
The requirement of efficient security protocols to protect confidential information from ever evolving cyber threat has been highlighted by the fast implementation of wireless sensor network (WSN) in numerous important domains, including healthcare monitoring, environmental surveillance and military operations. Despite their feasibility, traditional cryptographic approaches are utilized behind the computing capability of recent attackers, including the possibility of future quantum computer attacks. In order to revolutionize WSN security, this study proposes utilizing quantum cryptography and other concepts from quantum computing into the existing framework. This research leads to develop a communication protocol that is extremely safe and against the cyber-attacks. This protocol utilizes the quantum key distribution (QKD) to protect data and granting its confidential. In addition to improving security, the new architectures begin with change that makes WSN more capable of dependable and secure operations in environment where data protection is essential. A complete framework for protecting WSNs from new cyber threats in the future while keeping data transmissions and efficient process with reliability is discussed in this study. This proposed method enhances the dependencies in WSN installation across various industries, which is important for applications that need strong security measures. Evaluation results shows that the proposed method outperformed all existing methods in terms of relevant metrics.

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